ARCHITECTURE AND DECISION RECORD

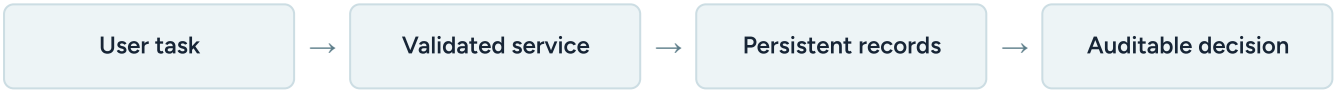
A network asset inventory and maintenance workbench

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Python | FastAPI | PostgreSQL | Asset reconciliation

Problem and contribution

This original design case study explores how to create an accountable equipment record that connects stable asset identity, authorized discovery observations, and maintenance history. I authored the requirements breakdown, architecture boundaries, data relationships, failure scenarios, and review rationale presented here. The design is a fictional presentation example, with no claim of a production deployment or measured client result.



Architecture decisions and tradeoffs

- Use an explicit administrator-approved address allowlist and rate limits for discovery workers.
- Separate observations from verified inventory so transient devices cannot silently become trusted records.
- Keep secrets in environment-managed configuration and never include raw credentials in export files or task logs.

Core records

Record	Responsibility
assets	The asset tag identifies equipment independently of a current IP address.
interfaces	One device may expose multiple interfaces and change its address over time.
address_observations	Observations are timestamped evidence, not an assumed permanent identity.
locations	Location transfers are logged rather than silently replacing the previous assignment.

The service layer owns authorization and business constraints. Interface validation improves the experience but cannot replace server checks. Event records preserve attributable decisions and their timing without logging unnecessary private content.

VERIFICATION AND REVIEW PRACTICE

Failure cases, evidence, and limitations

Proposed verification matrix

Scenario	Method	Expected behavior
Discovery boundary	Submit an address outside the approved lab range and a malformed target.	The worker rejects it before sending probes and records the rejected request.
Identity reconciliation	Import the same asset with changed IP addresses and two interfaces.	A reviewer can resolve identity without losing observation history or creating duplicate equipment.
Role separation	Attempt asset retirement and bulk export using a technician account.	Unauthorized actions are rejected on the server and appear in the audit log.
Maintenance lifecycle	Open, assign, close, and reopen a work order with attachment-free notes.	Each transition retains author, time, and reason; closed tasks are still readable.

Technical review

Keep discovery observations separate from verified inventory. Validate the allowed target range in the worker as well as the form, and avoid including raw topology credentials in logs or exported reports.

How to evaluate the design

Measure verified duplicate rate, successful reconciliation of changed addresses, rejected out-of-scope discovery attempts, work-order task success, and the age of unresolved observations. Do not equate a responding address with network health or an authorized owner.

The matrix specifies intended tests, not completed test results. A reviewer should capture the environment, initial records, inputs, expected state, observed state, and any unresolved defect before interpreting the outcome. Repeated runs should use separate unique request keys where relevant and preserve failures for analysis.

Scope limitations

No intrusive scanning, password collection, automated device reconfiguration, vulnerability exploitation, or claims of complete real-network coverage.

Check the trust boundary between automatically collected observations and the manually verified asset inventory.

> **Authorship and reuse**
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> Prepared by Adrian Castillo as an original presentation work sample. All narrative, schemas, and test scenarios are original illustrative material. There are no external client documents, employment certificates, real personal records, or fabricated citations in this case study.